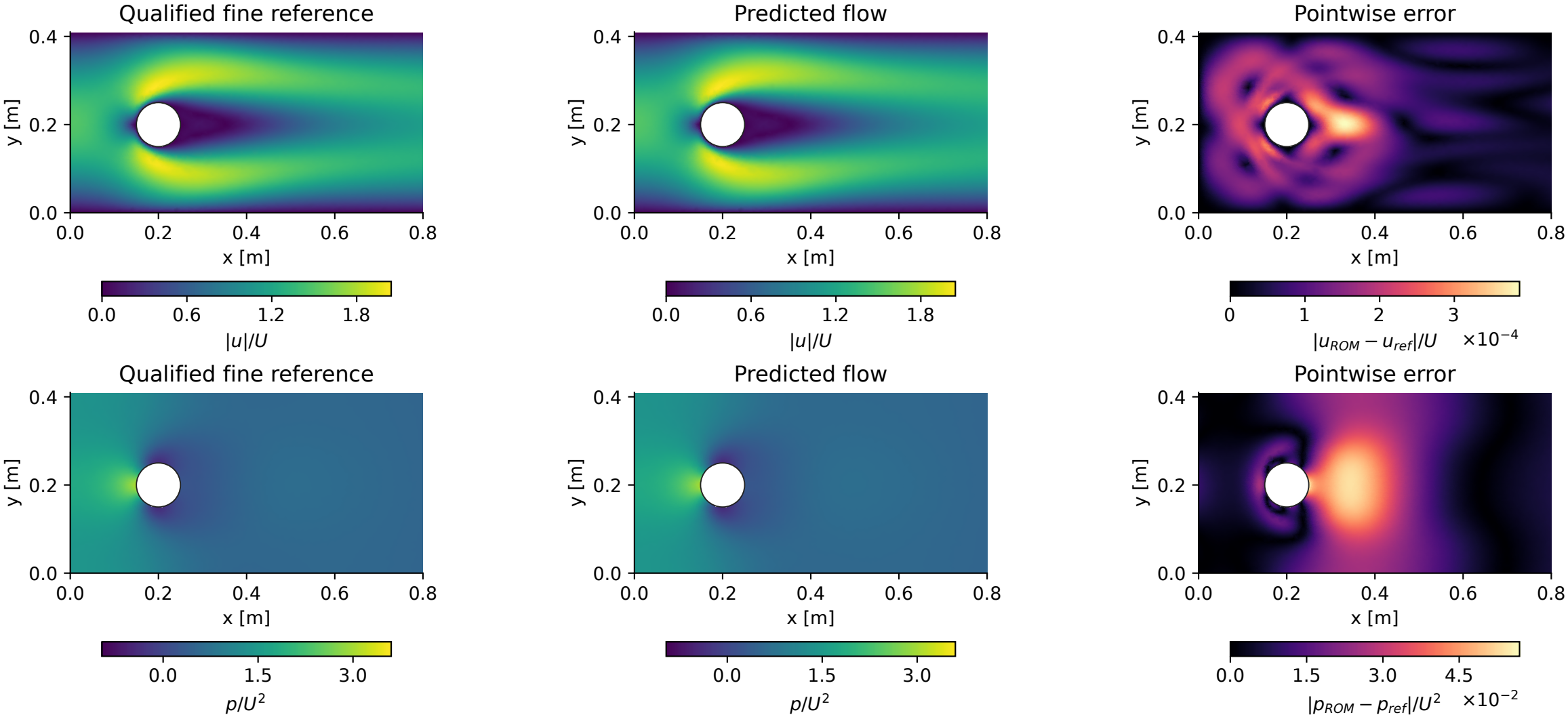

Standard POD–Galerkin: solved flow — Re = 23.125
Velocity (top) and pressure (bottom); near-cylinder view
Qualified fine reference
Predicted flow
Pointwise error
Qualified fine reference
Predicted flow
Pointwise error
y [m]
x [m]
0.0
0.2
0.4
0.6
0.8
|u|/U
0.0
0.6
1.2
1.8
|u_ROM − u_ref|/U
×10^−4
0
1
2
3
p/U^2
1.5
3.0
|p_ROM − p_ref|/U^2
×10^−2
4.5
Two POD velocity modes plus pressure-stability enrichment. Actual converged flow solve.
Display interpolation of saved samples; error scales differ between figures. Development cases, not validation.